TAA LEMMA 159: THE ARITHMETIC EQUATION OF STATE. PRIMON GAS AND BOST–CONNES, TIME AS IMAGINARY TEMPERATURE, $dT \wedge dS = dP \wedge dV$, CLAUSIUS VERSUS PASSIVITY, RUPPEINER–CONTACT GEOMETRY, AND LEE–YANG

J. W. MCGREEVY; FORMALIZATION BY GROK BOT

RH is not proved. Nothing here proves or claims to prove the Riemann Hypothesis. The lemma sorts thermodynamic statements about ζ into theorems, restatements of RH and analogies, and tests each against Davenport–Heilbronn (DH). Labels: [THEOREM], [RESTATEMENT] (equivalent to RH or to a known statement), [ANALOGY], [HEURISTIC], [NUMERICAL] (scripts `num/lemma159_*.py`, data `num/l159/checks.json`), [OPEN], [CONJECTURE]; sources [CITED] (opened for this lemma, or a local copy re-read), [BIB] (standard, not opened), [TAA-L] (earlier TAA lemma).

ABSTRACT. • **PV ↔ TS** (t727u; [THEOREM]/[NUMERICAL]). $0 = d(dU)$ gives $dT \wedge dS = dP \wedge dV$. So (P, V) and (T, S) are two Darboux charts of *one* area form, with Jacobian 1, and the four Maxwell relations are this one identity. J.W.'s "perfect symplectomorphism" is a classical theorem. Every cycle has work = heat = enclosed area, with the same orientation in both planes (checked for ideal and van der Waals gases: 1.272345 = 1.272345).

- **Cochains, not cohomology** ([THEOREM]). $P\,dV$ and $T\,dS$ are two primitives of one exact 2-form. Work over a cycle is the *flux* of that form, not a cohomology class, because the state space has no holes. The first law says $[\delta Q - \delta W] = 0$. Carathéodory's integrating factor $1/T$ is automatic in two dimensions; its content is universality, in three or more. The genuinely cohomological object on the arithmetic side is $d \log \xi$ on $\mathbb{C} \setminus \{\text{zeros}\}$: periods $2\pi i \cdot$multiplicity. These count zeros and are ghost-blind ([NUMERICAL]).
- **Clausius vs passivity** ([ANALOGY]/[RESTATEMENT]/[NUMERICAL]). Reading $\Gamma_\rho = 2|\beta - \frac{1}{2}|$ as entropy production gives an *empty* second law, since $\Gamma \geq 0$ by definition. RH is the *equality* $\Gamma \equiv 0$, i.e. reversibility. The positivity law that does work is passivity: RH $\Leftrightarrow \Re\, \xi'/\xi > 0$ on $\Re s > \frac{1}{2}$ (Lagarias/Hinkkanen). In that law a ghost is *gain*: ζ has minimum 0.0467, DH -15.08 at $(0.75, 85.7)$. On the line, Foster's monotone reactance is the Laguerre inequality, which RH implies but which does not imply RH.
- **The dictionary** (t726u; [THEOREM]/[ANALOGY]). $Z = \zeta(\beta)$; U, $S = \partial_T(T \log Z)$, $C = \beta^2 \mathrm{Var}(\log n)$ and $\Omega = -T \log \zeta$ are exact. The time term is t in $\mathrm{Tr}(e^{-\beta H} e^{itH}) = \zeta(\beta - it)$ (KMS). There is *no* volume, and the gas is not extensive, so "PV" is only a name for $-\Omega$. An energy cutoff gives an honest equation of state $P_X(\beta) = \sum_{n \leq X} \Lambda(n) n^{-\beta}/\log n$. Its smooth part comes from the pole (the L146 buoyancy/displaced reference); its fluctuation is governed by the zeros ([RESTATEMENT]).
- **Ghost test** ([THEOREM]/[NUMERICAL]). Gas positivities separate DH: its KMS characteristic function is not positive definite (Toeplitz -7.78); it is not infinitely divisible (466 negative Lévy coefficients); and its heat capacity is *negative* for $1 < \beta < 4.23183$. None of these forces RH. Epstein $x^2 + 5y^2$ has positive weights and an off-line zero; $\zeta(s)\zeta(s + a)$ is a free gas with zeros off its own FE line.
- **Geometry and Lee–Yang** ([CITED]/[THEOREM]). Ruppeiner $g = -\mathrm{Hess}\, S$, the contact structure (Mrugała), and a compatible J that *always* exists on a 2-dimensional state surface: the content of MP4 is commutation with a flow (L157/L158), and equilibrium has no flow. ζ's zeros are Fisher, not Lee–Yang, zeros. The exact Lee–Yang form is Newman's: RH $\Leftrightarrow \Lambda \leq 0$.

Date: 27 October 2026 (J.W. t726u, t727u; approved by card answer t727s2797).

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1. J.W.'s messages (verbatim) and the task

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t726u (J. W. McGreevy, verbatim):
Okay we had talked about density, half density, a mass like object and volume like object and then
↪  buoyancy... now to take this one step further - if we can identify a pressure term associated
↪  with this... and a time term... this might get us to a stage where every precise mathematical
↪  object would be definable in terms of time, pressure, volume, and entropy... if this is possible
↪  - we could reduce everything to precise absolute statements about thermodynamics.
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t727u (J. W. McGreevy, verbatim):
"Only gains you something where a positivity law does real work"...

Taking this literally - could the positivity law we are hoping to define not be understood as "work"
↪  on the system. I.e. something akin to a pressure-volume plane in a pressure volume loops? The PV
↪  loop itself would be analogous to cochains and cocycles. While if this is true the i think the
↪  "work" would be the cohomology theory... there is also the temperature-entropy plane which i
↪  have done work to suggest is a perfect symplectomorphism of the PV plane.
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Card t727s2797 (question: "Start L159 "The arithmetic equation of state" with the PV/TS cycle,
↪  Clausius-as-positivity and Lee-Yang sections?") -- J. W. McGreevy's answer, verbatim:
Yes, start L159 with the PV/TS and Clausius sections
```

Reading. The earlier TAA material t726u refers to is: *half density* (L145, the unitary twist $|x|^{1/2}$; atlas `half_density`), *density/volume* (L141 §7, area $\pi/3$ of the modular surface; `density_volume`), the *mass-like* object (L150 mass shell $\rho(1-\rho) = \frac{1}{4}+\gamma^2$; `mass_shell_even_xi`) and *buoyancy* (L146 §7: Archimedes as the subtraction of a displaced reference, $N^2 = -(g/\rho)\rho'$; `archimedes`, `mass_density`, `displaced_reference`). In t727u, "work" means $\oint P\,dV$; "cochains and cocycles" are read as simplicial/de Rham cochains on the state space; "perfect symplectomorphism of the PV plane" is read as the claim that $(P, V) \mapsto (T, S)$ preserves area and orientation. Earlier TAA thermodynamics is reused, not redone: L31 (BC, entropy), L44 (primon gas), L116 (Fisher/Ruppeiner/Mrugała, Laguerre), L118 (passivity), L124 (Fisher zeros, Lee–Yang, heat-flow ghosts).

Task (card answer). Start with the PV/TS and Clausius sections. (A) dictionary; (B) time = imaginary temperature; (C) PV/TS theorem, figure; (D) cochains vs cohomology, Carathéodory; (E) Clausius as a candidate positivity, Γ as entropy production, ζ vs DH numerics; (F) stability, Ruppeiner, contact geometry, the "missing J"; (G) Lee–Yang and de Bruijn–Newman; (H) ghost test throughout. Corrections C159.1–15 audit the chat claims.

Conventions. L103 ghost convention: a *ghost* is an off-line quartet $\{\rho, \bar\rho, 1-\rho, 1-\bar\rho\}$; $\Gamma_\rho = 2|\beta - \frac{1}{2}|$ (L154). DH $= \frac{1}{2}(1 - i\kappa)L(s,\chi) + \frac{1}{2}(1 + i\kappa)L(s,\bar\chi)$, χ mod 5, $\chi(2) = i$, $\kappa = 0.284079\ldots$ (L43/L141/L158). Units $k_B = 1$, primon energy scale $E = 1$. "Deninger's conjectured cohomology" means axioms MP0–MP4 (L151 Conj. 4.1); no new name is coined. Certified widths: L156 (`num/l156/`, via `num/l157/checks.json`).

2. (A) The dictionary: primon gas and Bost–Connes

Proposition 2.1 ([THEOREM]; [CITED] for the systems)**.** *Let $\mathcal{K}$ be the bosonic Fock space on $\ell^2(\{primes\})$, with basis $|n\rangle$, $n \geq 1$ (unique factorization), and $H|n\rangle = \log n\,|n\rangle$. For $\beta > 1$:*

(a) $Z(\beta) = \operatorname{Tr} e^{-\beta H} = \zeta(\beta) = \prod_p (1-p^{-\beta})^{-1}$. The sum runs over all occupation numbers at fugacity 1, so this is a grand-canonical partition function with $\mu = 0$, and $\Omega(\beta) := -T\log Z = \sum_p T\log(1-p^{-\beta})$ is a sum of independent single-mode grand potentials.

(b) $U = -\zeta'/\zeta(\beta) = \langle \log n\rangle_\beta$; $F = \Omega = -T\log\zeta$; $S = \partial_T(T\log Z) = \log\zeta(\beta) + \beta U$; $C = \beta^2(\log\zeta)''(\beta) = \beta^2 \mathrm{Var}_\beta(\log n) > 0$.

(c) The number of states with $E \leq E_0$ is $\lfloor e^{E_0} \rfloor$. This density of states e^E makes Z diverge for $\beta \leq 1$ (Hagedorn temperature $T_H = 1$).

Proof. (a) Expand each factor $(1-p^{-\beta})^{-1} = \sum_k p^{-k\beta}$ (one bosonic mode of energy $\log p$); multiply and use unique factorization. (b) Standard identities for $Z = \operatorname{Tr} e^{-\beta H}$: $U = -\partial_\beta \log Z$, $S = \beta U + \log Z$, $\partial_\beta^2 \log Z = \mathrm{Var}(H)$. (c) The states with $\log n \leq E_0$ are $n \leq e^{E_0}$. □

Sources ([CITED]). Wikipedia "Primon gas" (local copy `lit/lemma124/src`, re-read): $Z = \zeta(s)$, Hagedorn divergence at $s = 1$, Julia and Spector independently (Julia 1990, Springer Proc. Phys. 47, 276–293; Spector, CMP 127 (1990) 239–252; bibliographic data verified in L44, contents not opened). Bost–Connes (BC95, full IHES preprint `lit/lemma31/bc95.txt`, re-read): §1 defines KMS_β by a holomorphic $F_{x,y}$ on $0 < \Im z < \beta$ with $F(t) = \varphi(x\sigma_t(y))$ and $F(t+i\beta) = \varphi(\sigma_t(y)x)$, and records $\log \operatorname{Tr}(e^{-\beta H}) = \sup_\varphi(S(\varphi) - \beta\varphi(H))$. Theorem 5: for $0 < \beta < 1$ there is a unique KMS_β state (type III_1); for $\beta > 1$ the extreme KMS_β states are indexed by the embeddings $\mathbb{Q}^{\mathrm{cycl}} \hookrightarrow \mathbb{C}$, are of type I_∞, and have normalization $1/\zeta(\beta)$; "the critical temperature here is $T_c = 1$", and $\mathrm{Gal}(\mathbb{Q}^{\mathrm{cycl}}/\mathbb{Q})$ is spontaneously broken for $\beta > 1$.

Numbers ([NUMERICAL], 30 digits). $\beta = 2$: $\log Z = 0.497700$, $U = 0.569961$, $S = 1.637622$, $C = 3.537927$, $(\log\zeta)''(2) = 0.884482$ (L116: 0.884481833963524); $|S - \partial_T(T\log Z)| \leq 2\cdot 10^{-30}$ at $\beta = 3, 2, 1.5, 1.1, 1.01$. Near the Hagedorn point, $C(1.1) = 120.8$ and $C(1.01) = 10200.8$.

quantity	honest meaning for ζ	label	earlier TAA material
$T = 1/\beta$	$\beta = \Re s$	[THEOREM]	L38: $(\beta, t) = (\Re s, -\Im s)$
U, S, C, Ω	Prop. 2.1(b), $\beta > 1$	[THEOREM]	L39 entropy; L116 Fisher = C/β^2
density	density of states e^E (Hagedorn)	[THEOREM]	`density_volume` (area $\pi/3$): a different density, [ANALOGY]
half density	$\beta = \frac{1}{2}$: weights $n^{-1/2}$	[ANALOGY]	L145 twist $\lvert x\rvert^{1/2}$ (two $\frac{1}{2}$'s, C145)
mass-like	$\beta(1-\beta) = \frac{1}{4}$ at $\beta = \frac{1}{2}$; mass shell $\rho(1-\rho) = \frac{1}{4} + \gamma^2$	[RESTATEMENT]	L150
volume V	*none*; options V1–V3 below	[THEOREM] (absence)	L141/L146 cusp height, area
pressure P	*none*; $P_X(\beta)$ below is a definition	[RESTATEMENT] (def.)	L146 buoyancy = pole subtraction
time	t in $\operatorname{Tr}(e^{-\beta H}e^{itH}) = \zeta(\beta - it)$	[THEOREM]	§3

Proposition 2.2 ([THEOREM]/[NUMERICAL]: no volume, no extensivity)**.** *(V1)* Mode count. *Let $Z_X(\beta) = \prod_{p\leq X}(1-p^{-\beta})^{-1}$, with $\pi(X)$ modes. For $\beta > 1$, $\log Z_X \to \log\zeta(\beta)$ is bounded, so $\log Z_X/\pi(X) \to 0$. At $\beta = 1$, $\log Z_X(1) = \gamma + \log\log X + o(1)$ (Mertens' third theorem, [BIB]). The gas is not extensive in any mode count. Numbers ([NUMERICAL]): $\beta = 2$: $\log Z_X = 0.495882$, 0.497573, 0.497700 at $X = 10^2, 10^3, 10^7$ (limit 0.497700), per mode $7.5\cdot 10^{-7}$ at 10^7; $\beta = 1$, $X = 10^7$:* 3.357168 *against* $\gamma + \log\log X = 3.357158$. *(V2)* Energy cutoff. *Define, as a definition and not an*

identification, the arithmetic equation of state

$$P_X(\beta) := \sum_{n \le X} \frac{\Lambda(n)}{n^\beta \log n} = \sum_{p^k \le X} \frac{p^{-k\beta}}{k}, \qquad \mathrm{Li}_\beta(X) := \int_2^X \frac{u^{-\beta}\, du}{\log u}.$$

By partial summation against ψ and the explicit formula, $P_X(\beta) - \mathrm{Li}_\beta(X)$ equals a constant plus $-\sum_\rho \int^X u^{\rho-1-\beta} du/\log u$ plus lower-order terms. The smooth part Li_β comes from the pole at $s=1$ (the "displaced reference" of L146), and the oscillating part is carried by the zeros ([THEOREM], standard). RH $\Leftrightarrow$ the oscillation is $O(X^{1/2-\beta+\varepsilon})$ (von Koch type, [RESTATEMENT]/[BIB]). ([NUMERICAL]) $\beta = \frac{1}{2}$: $P_X - \mathrm{Li}_{1/2} = 0.268, 0.292, 0.290, 0.246, 0.247$ at $X = 10^3, \dots, 10^7$ (bounded, consistent with RH; not evidence). $\beta = 0.75$: $0.240 \to 0.241$ (convergent). (V3) Cusp height / area *(L141, L146). The truncated modular surface has a genuine volume $\pi/3 - 1/Y$, but it is the volume of a different object, not of the primon gas ([ANALOGY]).*

What t726u can and cannot get. Temperature, energy, entropy, heat capacity and free energy are exact (Prop. 2.1). The *time term* exists, with the KMS precision of §3. A *pressure term* exists only after choosing a cutoff (V2), and then its fluctuating part is the zeros; that is a restatement, not new information. "Every precise mathematical object definable in T, P, V, S" cannot hold as a theorem. All thermodynamic potentials are Legendre transforms of one generating function on one Legendrian surface (§7), so the dictionary is a change of variables. It gains something only where a positivity law holds (§6, §9). **Buoyancy, precisely** ([ANALOGY]). L146 read Archimedes as weight relative to displaced fluid, i.e. subtraction of a reference. In (V2), the reference is $\mathrm{Li}_\beta(X)$, the contribution of the pole, and the "buoyant" remainder is the zero sum. The same subtraction is $\psi(x) - x$ (`psi_minus_x`).

3. (B) Time as imaginary temperature

Proposition 3.1 ([THEOREM])**.** *In the primon/BC system, $\sigma_t(\mu_n) = n^{it}\mu_n$ (BC95 §5 eq. (2), re-read), i.e. $\sigma_t = \mathrm{Ad}\, e^{itH}$. For $\beta > 1$: (a) $\mathrm{Tr}(e^{-\beta H} e^{itH}) = \sum_n n^{-\beta+it} = \zeta(\beta - it)$. Thus $\varphi_\beta(t) := \zeta(\beta - it)/\zeta(\beta) = \langle e^{itH} \rangle_\beta$ is the characteristic function of the energy $\log n$ under the Gibbs measure. It is positive definite (Bochner) and nowhere zero (Euler product). (b) $\log \varphi_\beta(t) = \sum_{n \ge 2} \frac{\Lambda(n)}{\log n} n^{-\beta}(n^{it} - 1)$ has nonnegative weights, so the zeta distribution is infinitely divisible (compound Poisson): a* free *gas. (c) The KMS condition continues correlations into the strip $0 < \Im z < \beta$ (BC95 §1). Complex time $t + i\beta'$ and inverse temperature are the same complex variable only in this sense: $e^{-\beta H} e^{itH} = e^{-(\beta - it)H}$.*

Proof. (a) Absolute convergence; $\varphi_\beta(t) = \sum_n \frac{n^{-\beta}}{\zeta(\beta)} e^{it \log n}$ is the Fourier transform of a probability measure. (b) Take the log of the Euler product, $-\log(1 - p^{-s}) = \sum_k p^{-ks}/k$. (c) is the definition. □

The critical line ([RESTATEMENT], with a caveat). Writing $s = \beta - it$ makes $\Re s = \frac{1}{2}$ the line $\beta = \frac{1}{2}$ of complex temperature. That line lies in the Hagedorn-divergent region: $\mathrm{Tr}\, e^{-\beta H} = \infty$ for $\beta \le 1$, so $\zeta(\frac{1}{2} - it)$ is an analytic continuation, not a trace. BC's $\mathrm{KMS}_{1/2}$ state exists and is unique (Thm. 5(a)), but its normalization is not $\zeta(\frac{1}{2})$ (C116.2: "BC high-T phase $\neq$ critical strip"). So "the critical line is a line of complex temperature" is a restatement of $\Re s = \frac{1}{2}$; no thermodynamic principle singles out $\beta = \frac{1}{2}$ (L116: BC distinguishes only the pole $\beta = 1$).

Ghost test of the time layer ([NUMERICAL], `checks.json` B). Toeplitz matrix $[\varphi(t_i - t_j)]$ with $\beta = 2$, $t_j = 0.25j$, $j < 120$: ζ has minimum eigenvalue $1.3 \cdot 10^{-10}$ (PSD up to rounding). DH has minimum -7.78, with 10 negative eigenvalues: DH's "Gibbs weights" $(1, \kappa, -\kappa, -1, 0)$ mod 5 are signed, so it has no KMS state and is not a gas. Lévy coefficients $b(n)$ of $\log f$ for $n \le 2000$: ζ, 0 negative; DH, 466 negative and 615 nonzero off prime powers (an "interacting" gas); $\zeta(s)\zeta(s + \frac{1}{2})$, 0 negative; Epstein $x^2 + 5y^2$ (weights $r(n)/2 \ge 0$), 40 negative. The time layer is **DH-sensitive**. It is not RH-forcing (§9).

4. (C) The PV/TS theorem: J.W.'s symplectomorphism

Theorem 4.1 ([THEOREM]; J.W. t727u)**.** *Let Σ be the 2-dimensional state surface of a simple system (fixed N), with $dU = T\,dS - P\,dV$. Then*

(a) $\omega := dT \wedge dS = dP \wedge dV$ on Σ. Wherever (P, V) and (T, S) are both coordinates, $\partial(T, S)/\partial(P, V) = 1$: the change of chart is area- and orientation-preserving, hence a symplectomorphism of 2-dimensional symplectic manifolds.

(b) The four Maxwell relations are (a) written in the charts $(x, y) = (P, S), (T, V), (P, T), (S, V)$: $\partial(T, S)/\partial(x, y) = \partial(P, V)/\partial(x, y)$. Conversely, any one of them in a coordinate chart implies (a). The equivalence "Maxwell $\Leftrightarrow$ Jacobian 1" holds.

(c) For every region $D \subset \Sigma$, $\iint_D dP \wedge dV = \iint_D dT \wedge dS$. For a cycle $c = \partial D$, $W = \oint_c P\,dV = \oint_c T\,dS = Q$, the signed enclosed area in either chart. An engine cycle (positive W) runs clockwise in the (V, P) diagram, V horizontal, and its image runs clockwise in the (S, T) diagram, S horizontal.

Proof. (a) $0 = d(dU) = dT \wedge dS - dP \wedge dV$. (b) Divide (a) by $dx \wedge dy$; e.g. $(x, y) = (P, S)$ gives $(\partial T/\partial P)_S = (\partial V/\partial S)_P$. A Maxwell relation in a chart (x, y) says that ω_{TS} and ω_{PV} have the same coefficient on $dx \wedge dy$, and that is (a). (c) Stokes: $\oint_{\partial D} P\,dV = \iint_D dP \wedge dV$ and likewise for $T\,dS$. $W = Q$ is also $\oint dU = 0$. Orientation: $dT \wedge dS = dP \wedge dV$ means the charts (S, T) and (V, P) induce the same orientation, since $dS \wedge dT = -\omega = dV \wedge dP$. □

Source ([CITED]). Wikipedia "Maxwell relations", §"Derivation in terms of Jacobians" (fetched 2026-10-27): "Taking the cycle infinitesimal yields $\partial(P, V)/\partial(T, S) = 1$. That is, the map preserves area. Maxwell's relations therefore state in a cyclic process, that the work done by the system ... must be equal to the heat absorbed". The source there is Ritchie, Am. J. Phys. 36 (1968) 760, not opened. Precision (C159.9): it is *one* 2-form in two charts. It fails where (P, V) or (T, S) stop being coordinates, e.g. on phase-coexistence lines, where P and T are dependent.

Numbers ([NUMERICAL], `checks.json` C). Monatomic gas with $Nk = 1$: $U = \frac{3}{2}T - a/V$, $P = T/(V - b) - a/V^2$, $S = \log(V - b) + \frac{3}{2}\log T$. At 200 random points, the ideal ($a = b = 0$) and van der Waals ($a = 1$, $b = 0.2$) Jacobians lie in $1 \pm 2 \cdot 10^{-9}$. Ellipse cycle $V = 2.5 + 0.9\cos\theta$, $P = 1.2 - 0.45\sin\theta$ (clockwise): $W = 1.272345024$ and $Q = 1.272345025$ (ideal and vdW), against $\pi \cdot 0.9 \cdot 0.45 = 1.272345025$. Both shoelace signed areas are -1.272345, so both are clockwise and the orientation is preserved. Maxwell $(\partial T/\partial P)_S = (\partial V/\partial S)_P = 0.8$ (ideal) and 0.9 (vdW) at $(V, P) = (2, 1.1)$. Carnot ($T_h = 2$, $T_c = 1$): $\sum Q/T = 0$, efficiency $0.5 = 1 - T_c/T_h$.

Arithmetic counterpart? ([OPEN]). The primon gas has no V (Prop. 2.2), so Theorem 4.1 has no direct instance for ζ. With the cutoff (V2), the pair $(\beta, \log X)$ gives a 2-parameter family $\log Z_X(\beta)$, but no first law has been derived for it. Whether (T, S) and $(P_X, \log X)$ share a 2-form for ζ is open. Even if they did, Theorem 4.1 has no arithmetic input and is ghost-blind.

5. (D) Cochains, cocycles and cohomology; Carathéodory

Proposition 5.1 ([THEOREM])**.** *On the state surface Σ (contractible, e.g. an open quadrant): (a) Integration gives 1-cochains $c \mapsto \int_c P\,dV$ and $c \mapsto \int_c T\,dS$ (de Rham map). Neither is a cocycle: $d(P\,dV) = d(T\,dS) = \omega \neq 0$. They are two primitives of the same exact 2-form, and they differ by the exact form dU (first law). (b) $H^1(\Sigma) = 0$, so every 1-cycle bounds and $\oint_c P\,dV = \iint_D \omega$ is the* flux *of ω through D. Work is therefore not a cohomology class. The first law is the statement $[\,T\,dS - P\,dV\,] = 0$, which is automatic on a contractible Σ once U exists. (c)* Carathéodory. *$\delta Q = dU + P\,dV$ is not closed, but $\delta Q/T = dS$ is exact. On a 2-dimensional Σ every nowhere-zero 1-form has a local integrating factor, because the Frobenius condition $\theta \wedge d\theta = 0$ is vacuous. The content of Carathéodory's principle is in dimension ≥ 3 (several work coordinates, composite systems), where it gives integrability and an integrating factor that is a universal function of empirical temperature.*

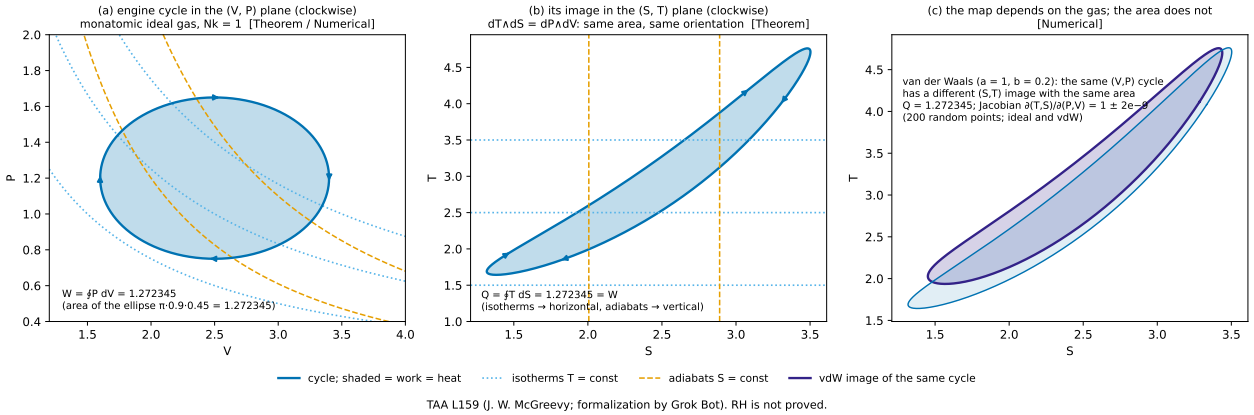


FIGURE 1. Theorem 4.1. (a) An engine cycle in the (V, P) plane; (b) its image in the (S, T) plane, where isotherms are horizontal and adiabats vertical; same shaded area 1.272345, same (clockwise) orientation. (c) A van der Waals gas sends the same (V, P) cycle to a different (S, T) loop of the same area. Colours: one per concept (CONCEPT_COLORS); [THEOREM]/[NUMERICAL].

Proof. (a),(b) Stokes and the Poincaré lemma. (c) In 2D, $\theta \wedge d\theta$ is a 3-form, hence 0, so $\ker \theta$ is integrable and $\theta = f\, dg$ locally. In dimension ≥ 3 the condition is a real constraint ([BIB]). □

Source ([CITED]). Wikipedia "Second law of thermodynamics", §Principle of Carathéodory (fetched 2026-10-27): "In every neighborhood of any state S of an adiabatically enclosed system there are states inaccessible from S"; from it, "$\delta Q = TdS$"; and, importantly, Carathéodory's principle "needs to be supplemented by Planck's principle" to give the whole second law. Carathéodory 1909 (Math. Ann. 67) was not opened.

Name collision (C159.14). L157 Thm. 6.2 ("contraction ⇒ Carathéodory") refers to the Carathéodory *function class* (holomorphic, positive real part; Carathéodory–Toeplitz). It is unrelated to Carathéodory's thermodynamic principle; the two share only the name.

Proposition 5.2 ([THEOREM]/[NUMERICAL]: where the cohomology actually is)**.** *Let* Λ *be* ξ *or the completed DH function. Then* $d\log\Lambda = (\Lambda'/\Lambda)\, ds$ *is closed on* $\mathbb{C} \setminus \{zeros\}$*, and its period around a small loop at a zero of multiplicity* m *is* $2\pi i\, m$*. Its real part* $d\log|\Lambda|$ *is exact (with potential* $\log|\Lambda|$*, harmonic off the zeros; cf. L122 "Fisher charges"). The cohomology class therefore records* how many *zeros a loop encloses, not* where *they sit. [NUMERICAL]:* $\frac{1}{2\pi i}\oint d\log\Lambda = 1.0$ *for* ζ *on* $[0.2, 0.8] \times [13.5, 14.8]$*,* 1.0 *for DH on* $[0.7, 0.9] \times [85.5, 85.9]$ *(the ghost* ρ_1 *alone), and* 2.0 *on* $[0.1, 0.9] \times [85.5, 85.9]$ *(*ρ_1 *and* $1 - \bar{\rho}_1$*);* $|\mathrm{DH}(\rho_1)| = 3 \cdot 10^{-20}$.

Reading of t727u ([RESTATEMENT]/[ANALOGY]). "The PV loop is a cochain": correct, and specifically a non-closed 1-cochain whose coboundary is ω. "The work would be the cohomology theory": on a contractible state space work is a flux, not a class. The thermodynamic object that *is* a cohomology class is the free-energy form $d\log Z$ on the complex-temperature plane minus its Fisher zeros. Its periods are quantized $(2\pi i)$ and ghost-blind: an off-line zero has the same period as an on-line one (L152: off-line zeros show only as missing sign changes against $N(T)$). This matches L144's "cocycles give structure, not sign" (`cocycle_principle`). A ghost-sensitive statement needs a *sign* (§6), not a period.

6. (E) Clausius as the candidate positivity law; Γ as entropy production

Proposition 6.1 ([CITED]/[THEOREM])**.** *(a) Clausius: for a system exchanging heat* δQ *with reservoirs at temperature* T_{res} *over a cycle,* $\oint \delta Q / T_{\mathrm{res}} \leq 0$*, with equality for reversible cycles; equivalently, the reservoir entropy does not decrease (Wikipedia "Clausius theorem", fetched 2026-10-27).* T *here*

is the reservoir *temperature; the system temperature need not exist during an irreversible process. (b) [NUMERICAL]: Carnot $T_h = 2$, $T_c = 1$: $Q_h/T_h - Q_c/T_c = 0$. Conduction of $Q = 1$ from $T = 2$ to $T = 1$ produces entropy $\frac{1}{2}$.*

The chat analogy, tested ([ANALOGY], C159.12). The chat proposed reading $\Gamma_\rho = 2|\beta - \frac{1}{2}|$ as entropy production: $\Gamma = 0$ for a reversible mode and $\Gamma > 0$ for a dissipative one, so RH says "the arithmetic gas runs reversibly on every mode".

(i) The inequality is empty. $\Gamma \geq 0$ holds by definition, so a "second law" $\Gamma \geq 0$ carries no information. RH is the *equality* $\Gamma \equiv 0$, a reversibility (lossless) statement. This is the same conclusion as L118 Thm. 4.1 (RH = the undoubled system is lossless, with no arrow of time).

(ii) No zero-free entropy-production functional is known. A precise version would be a functional $\sigma[f] \geq 0$ built from Euler-product data without the zeros (requirement R5 of L158), with $\sigma = 0 \Leftrightarrow \Gamma \equiv 0$. $\sum_\rho w(\gamma)\Gamma_\rho$ is circular. None is known ([OPEN]).

(iii) The working positivity law has the *opposite* sign picture ([RESTATEMENT]). Hadamard: $\Re\frac{\xi'}{\xi}(\sigma + it) = \sum_\rho \frac{\sigma-\beta}{(\sigma-\beta)^2+(t-\gamma)^2}$, a Lorentzian of sign $\mathrm{sgn}(\sigma - \beta)$ per zero (L146/L154). Lagarias/Hinkkanen: RH $\Leftrightarrow \Re\,\xi'/\xi > 0$ on $\Re s > \frac{1}{2}$. This is passivity, i.e. nonnegative absorbed power (atlas `passivity`). In this law a ghost is *gain* (negative absorption) for $\frac{1}{2} < \sigma < \beta$: a local violation of the second law, not extra dissipation.

(iv) On the line the medium is purely reactive. $\xi'/\xi(\frac{1}{2} + it) = -i\,\Xi'/\Xi(t)$ is purely imaginary (lossless), and Foster's reactance theorem (Wikipedia, fetched: the reactance of a passive lossless one-port strictly increases with frequency) becomes $\frac{d}{dt}(-\Xi'/\Xi) = (\Xi'^2 - \Xi\Xi'')/\Xi^2 > 0$, the Laguerre inequality ([THEOREM]). RH $\Rightarrow$ Laguerre; Laguerre $\not\Rightarrow$ RH, since L116 found planted ghosts that are locally Laguerre-invisible ([TAA-L]).

(v) In de Bruijn–Newman the chat's sign is right. The forward (dissipative) heat flow *creates* off-line pairs; increasing t is the backward heat flow, which makes zeros real (C158.16). So "ghost = dissipation" is right in heat-flow time and wrong in the passivity picture. Any use of the analogy must say which one is meant.

Numbers ([NUMERICAL]; certified widths from L156, read from `num/l157/checks.json`). ζ: 52 zeros with $2 < t < 150$, complete and certified, all $\Gamma = 0$. DH: 122 on-line zeros below $t = 200$ ($\Gamma = 0$) and four off-line pairs, $\Gamma(\rho_1) = 0.617034364913$ ($\gamma = 85.699$), $\Gamma(\rho_2) = 0.301660161219$ (114.163), $\Gamma(\rho_3) = 0.148712100902$ (166.479), $\Gamma(\rho_4) = 0.448515389254$ (176.702); $\sum\Gamma = 1.516$ over the upper-half representatives. Passivity scan $t \in [80, 120]$ (step 0.1), $\sigma \in \{0.55, 0.65, 0.75, 0.85\}$: ζ minimum 0.0467 (at $\sigma = 0.55$, $t = 90.7$), never negative. DH minimum -15.08 at $(0.75, 85.7)$; at $\sigma = 0.55$ negative exactly on $[85.5, 85.9]$ and $[114.1, 114.3]$, around the two ghosts; at $\sigma = 0.85 > \beta_1$ positive (minimum 0.740). L146's finer grid gave -66.84 at $(0.794, 85.70)$ ([TAA-L]). FE check on the line: $|\Re\Lambda'/\Lambda(\frac{1}{2} + 100.3i)| = 0$ (ζ) and $4 \cdot 10^{-16}$ (DH).

7. (F) Stability, Ruppeiner geometry, contact geometry, and the "missing J"

Proposition 7.1 ([THEOREM]; [CITED] for the definitions)**.** *(a)* Stability = convexity. *For the canonical/grand-canonical gas, $C = \beta^2\partial_\beta^2 \log Z \geq 0$. For any Dirichlet series with $a(n) \geq 0$, $\log f$ is convex on the half-line of absolute convergence (Cauchy–Schwarz, $\partial_\beta^2 \log f = \mathrm{Var}(\log n)$). In the energy representation stability means that S is concave, equivalently that $U(S, V)$ is convex. (b)* Metrics *(Wikipedia "Ruppeiner geometry", fetched): $g_R = -\mathrm{Hess}\,S$ (positive by stability), Weinhold $g_W = \mathrm{Hess}\,U$, $ds_R^2 = ds_W^2/T$. g_R is flat for non-interacting systems, and it is Lorentzian when the heat capacity is negative (Reissner–Nordström example). (c)* The compatible J is free in 2D. *On an oriented 2-dimensional Σ with a metric g and an area form $\omega = f\,\mathrm{vol}_g$, $f > 0$, rotation by $90°$ in g gives J with $J^2 = -1$, $g(J\cdot, J\cdot) = g$ and $\omega(\cdot, J\cdot) = f\,g$. So (ω, J, g) is a Kähler triple up to the conformal factor f.*

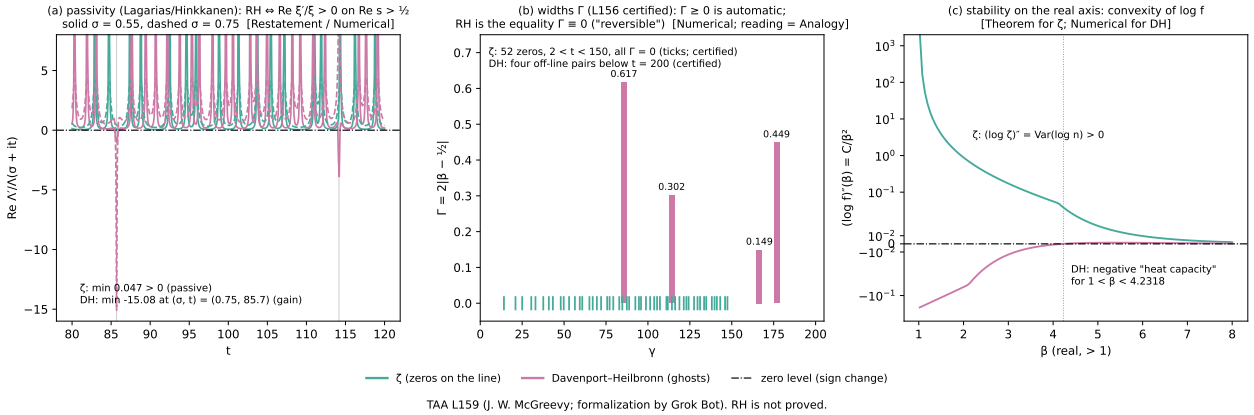


FIGURE 2. Three thermodynamic layers, ζ (teal) against DH (purple). (a) Passivity $\Re\Lambda'/\Lambda$ at $\sigma = 0.55$ (solid) and 0.75 (dashed): DH shows gain at its ghosts ([RESTATEMENT]/[NUMERICAL]). (b) Certified widths: $\Gamma \geq 0$ is automatic, RH is $\Gamma \equiv 0$ ([NUMERICAL]; reading [ANALOGY]). (c) Real-axis stability $(\log f)'' = C/\beta^2$: positive for ζ (Theorem), negative for DH on $1 < \beta < 4.23183$ ([NUMERICAL]).

Proof. (a) Hölder. (b) Definitions. (c) In 2D every metric has a unique orientation-compatible rotation by 90°, and every 2-form is a multiple of vol_g. □

Numbers ([NUMERICAL], sympy + numpy, `checks.json` F). Ideal gas $S = \frac{3}{2}\log U + \log V$ at $(U, V) = (3, 2)$: $g_R = \mathrm{diag}(1/6, 1/4)$, $\omega = dT \wedge dS = dP \wedge dV = \frac{1}{3}\, dU \wedge dV$, $J^2 + 1 = 10^{-16}$, $g(J\cdot, J\cdot) - g = 6 \cdot 10^{-17}$, Gaussian curvature 0, and $\omega/\mathrm{vol}_{g_W} = \sqrt{2/3}$ *identically* (sympy: $\sqrt{6}/3$). Van der Waals $(a, b) = (1, 0.2)$: curvature 0.2244 (interaction), $J^2 + 1 = 2 \cdot 10^{-16}$. For the primon gas, L116 found that the multiplicative (Euler) field is separable, hence Ruppeiner-flat, while the χ_4 field is curved with both signs. Curvature detects non-multiplicativity, not zeros ([TAA-L]).

So nothing is missing in the thermodynamic J (C159.13). The chat said the missing piece is "a compatible J, as in MP4". On a simple system's 2-dimensional state surface such a J always exists (Prop. 7.1(c)). This is the thermodynamic twin of L158's finding that positivity of $\mathrm{tr}(h \cup *h)$ is automatic for surface leaves. In both cases the content of MP4 is the *commutation* of J with a flow, $[J, \Theta] = 0$ (L157 Thm. 2.4, L158 Cor. 4.2). Equilibrium thermodynamics has no flow whose spectrum could be the zeros, so it has no MP4 analogue yet ([OPEN]).

Contact geometry ([CITED]). Mrugała 2000 (RIMS Kôkyûroku 1142, 167–181; local copy, reread): the thermodynamic phase space is a $(2n + 1)$-dimensional contact manifold, the equilibrium states are Legendre submanifolds, the contact structure "is related to the first law" and the metric structure "to the second law". His metric $G = dp_i\, dx^i + \theta \otimes \theta$ has signature $(n + 1, n)$, and only its restriction to a Legendre submanifold is the positive Fisher/Ruppeiner metric (L116, C116.1). Gibbs and Hermann are cited only through Mrugała. *Legendre and stability* ([THEOREM], L153). The potentials $U, F = U - TS, H, G$ are Legendre transforms of one generating function, and the Legendre transform of a quadratic form is finite iff the form is definite (L153 Legendre window; Fenchel–Moreau). Thermodynamic stability is exactly what makes the dictionary invertible. Where convexity fails, the transform returns the convex hull: Maxwell's equal-area construction ([BIB], not opened).

The ghost side, thermodynamically ([NUMERICAL]/[ANALOGY]). DH is positive on $(1, 10]$ but $(\log \mathrm{DH})'' < 0$ for $1 < \beta < 4.23183$ (`checks.json` P; e.g. -0.2037 at $\beta = 1.05$, -0.0585 at 2, $+0.0013$ at 5). Read as a heat capacity, DH has *negative* heat capacity near the Hagedorn point, the signature that Wikipedia's Ruppeiner article attaches to a Lorentzian thermodynamic metric ([ANALOGY]). This refines L116's "(S1) real-axis Hölder convexity unconditional and blind" (C159.15). Convexity

is unconditional only for positive weights; it fails for DH, but it holds for Epstein $x^2 + 5y^2$, which has an off-line zero, so it is DH-sensitive and not RH-forcing.

8. (G) Lee–Yang zeros, Fisher zeros and de Bruijn–Newman

Proposition 8.1 ([CITED]/[THEOREM]). *(a)* Lee–Yang *(1952; via Newman–Wu 2019 §2.5 and Wikipedia "Lee–Yang theorem/theory", local copies re-read). For a ferromagnetic Ising model ($J_{jk} \geq 0$), all zeros of Z in the fugacity $z = e^{-2\beta h}$ lie on $|z| = 1$. Newman (1974; NW19 Thm. 11) extends this to measures with the Lee–Yang property. Zeros in a field variable are* Lee–Yang *zeros; zeros in the inverse temperature are* Fisher *zeros. (b)* ζ's zeros are Fisher zeros, not Lee–Yang zeros *([THEOREM]; C124). With prime fugacities z_p, $\prod_{p \leq X}(1 - z_p p^{-\beta})^{-1}$ has no zeros at all. The zeros of $\zeta(\beta - it)$ are zeros in complex β of the continued partition function. (c)* The exact Lee–Yang form *([CITED]/[RESTATEMENT]). With Φ the Pólya kernel and $H_\lambda(z) = \int e^{\lambda u^2} \Phi(u) \cos(zu)\, du$, NW19 Thm. 9 (de Bruijn 1950, Newman 1976) gives $\Lambda \in (-\infty, \frac{1}{2}]$ such that H_λ has only real zeros iff $\lambda \geq \Lambda$, and RH $\Leftrightarrow \Lambda \leq 0$. Rodgers–Tao give $\Lambda \geq 0$ and Polymath $\Lambda \leq 0.22$ (L124/L150/L158, opened there). Increasing λ is the* backward *heat flow (C158.16): the factor $e^{\lambda u^2}$ is a Gaussian tilt of the kind Newman's theorem allows. RH says Φ already has the Lee–Yang-type property at $\lambda = 0$.*

Newman's motivation ([CITED]). NW19 p. 10, on the 1976 theorem: "the original motivation for this result was provided more by statistical physics and Euclidean field theory, as discussed in Section 2.5, than by analytic number theory." This confirms the chat claim that Newman's de Bruijn–Newman work grew out of the Lee–Yang setting (C159.5).

Toy ([NUMERICAL]). Three spins with uniform coupling J: ferromagnet $J = 0.4$, all three roots have $|z| = 1$ (to $2 \cdot 10^{-15}$); antiferromagnet $J = -0.4$, $|z| = 13.79, 1.00, 0.0725$. The positivity of the couplings is what does the work.

What would a Lee–Yang theorem for the prime gas be? ([OPEN]). It would need a field variable in which the zeros of ζ appear, plus a "ferromagnetic" positivity of couplings proved from the primes. (b) rules out prime fugacities. The only known reformulation, (c), is RH itself. L124 Direction "genuine field variable (characters)" (0.48) remains open. A character twist χ is a natural field, but it changes $L(s, \chi)$ rather than moving zeros of ζ, and DH is exactly a non-multiplicative mixture of two such sectors (L158).

Ghost world ([TAA-L], L124). For DH-type ghosts, $0 < \Lambda \leq 2\delta^2_{\max}$ (de Bruijn Thm. 3.2 via Polymath; $z = 2\gamma - 2i\delta$). An isolated pair heals at $t^* = 2\delta^2$, and neighbours accelerate the healing. Positivity of Λ is ghost-sensitive, but $\Lambda \leq 0$ is a restatement of RH.

9. (H) The ghost test, layer by layer

"DH-sensitive" means the layer distinguishes ζ from DH. "RH-forcing" means the property implies RH. Controls with ghosts relative to their own functional equation: Epstein $Z_{x^2+5y^2}$, with positive weights $r(n)/2$, FE $s \leftrightarrow 1 - s$ and an off-line zero $0.9329696975 + 15.6682495313i$ (L83/L153); $\zeta(s)\zeta(s + a)$, $0 < a < \frac{1}{2}$, with an Euler product, positive and infinitely divisible weights, and zeros on $\Re s = \frac{1}{2}$ and $\frac{1}{2} - a$, off its FE line $\Re s = \frac{1-a}{2}$ ([THEOREM]); Hurwitz $\zeta(s, a)$, $a \neq \frac{1}{2}, 1$, with positive weights and zeros in $1 < \sigma < 1 + \varepsilon$ (Davenport–Heilbronn 1936, Cassels 1961, via Wikipedia "Hurwitz zeta function", fetched; originals not opened).

layer	DH	RH-forcing?	note
PV/TS area identity (Thm. 4.1)	n/a	no	no arithmetic input: **ghost-blind**
periods of $d\log\Lambda$ (Prop. 5.2)	same ($2\pi i$)	no	counts zeros: **ghost-blind**
positive Gibbs weights; PD KMS characteristic function	fails (-7.78)	no	Epstein, Hurwitz pass: **DH-sensitive only**
real-axis convexity, $C \geq 0$	fails on $(1, 4.23)$	no	Epstein passes; refines L116 (C159.15)
infinite divisibility (free gas = Euler product)	fails (466 neg.)	no	$\zeta(s)\zeta(s+a)$ passes: **DH-sensitive only**
Ruppeiner flatness	curved? (cf. L116 χ_4; [ANALOGY])	no	measures non-multiplicativity
2D Kähler triple (ω, J, g)	exists	no	automatic (Prop. 7.1(c)): **blind**
Clausius: $\Gamma \geq 0$	holds	no	automatic: **blind**
reversibility $\Gamma \equiv 0$	fails $(0.617, \dots)$	$\Leftrightarrow$ RH	a functional of the zeros: circular (R5)
passivity $\Re\,\xi'/\xi > 0$, $\Re s > \frac{1}{2}$	fails (-15.08)	$\Leftrightarrow$ RH	[RESTATEMENT]; the working "second law"
Foster/Laguerre on the line	fails at ghosts	no ($\Leftarrow$ RH)	L116: locally invisible ghosts
Lee–Yang form, $\Lambda \leq 0$	$\Lambda_{\mathrm{DH}} > 0$	$\Leftrightarrow$ RH	[RESTATEMENT] (Newman, de Bruijn)

Verdict ([THEOREM] for the table entries, [HEURISTIC] for the moral). The thermodynamic layer has two kinds of positivity. (i) *Gas positivities* (positive weights, PD KMS function, convexity, infinite divisibility) hold for ζ and fail for DH, so they separate DH. They are *not* ghost-sensitive in general: each is satisfied by a control that has ghosts relative to its own FE line. (ii) Positivities *equivalent* to RH (passivity, $\Lambda \leq 0$, $\Gamma \equiv 0$) are restatements. No thermodynamic positivity in between is known. The chat claim "the thermodynamic layer is ghost-sensitive by construction" is corrected to "DH-sensitive by construction, not RH-forcing" (C159.7). What the controls show is missing: the gas positivity has to be coupled to the self-dual FE line of a degree-one object. That is the MP4 gap again (positivity tied to the functional-equation symmetry), not a new piece.

10. Corrections: audit of the 13 chat claims

C159.1 *Primon gas / Bost–Connes.* Correct ([CITED]). Precise form (BC95 Thm. 5): $Z = \zeta(\beta)$, a unique KMS state for $0 < \beta < 1$ (type III_1; §7 of BC95 proves uniqueness on $]0, 1]$), and for $\beta > 1$ extremal states indexed by embeddings of $\mathbb{Q}^{\mathrm{cycl}}$ (type I_∞) with normalisation $1/\zeta(\beta)$. Symmetry breaking with $T_c = 1$. The bare primon gas has a Hagedorn divergence at $T_H = 1$.

C159.2 *"Pressure" and "volume".* Corrected. The gas is grand canonical with $\mu = 0$ and has *no volume* and no extensivity. "$PV = T\log Z$" only names $-\Omega$. The independent modes have grand potentials, not pressures. The honest equation of state is $P_X(\beta)$ in the cutoff X, with the pole $\mathrm{Li}_\beta(X)$ as the displaced reference (buoyancy, Prop. 2.2).

C159.3 $S = \log\zeta + \beta U$. Correct ([THEOREM]; residual $\leq 2 \cdot 10^{-30}$).

C159.4 *Time as imaginary temperature.* Correct ([THEOREM]): $\mathrm{Tr}(e^{-\beta H}e^{itH}) = \zeta(\beta - it)$. Caveat: the critical line $\beta = \frac{1}{2}$ lies outside the trace-class region (C116.2), and $\zeta(\frac{1}{2} + it)$ is not a partition function.

C159.5 *Zeros = Lee–Yang zeros.* Corrected. They are Fisher zeros (complex β), not Lee–Yang zeros (Prop. 8.1(b)). The Lee–Yang circle theorem and Newman's statistical-physics motivation are confirmed (NW19). The exact statement is RH $\Leftrightarrow \Lambda \leq 0$.

C159.6 *Stability $C \geq 0$.* Correct for positive weights. Stability is concavity of S, and $C \to \infty$ as $\beta \to 1^+$ ($C(1.01) = 10200.8$). Controls (Epstein, $\zeta\zeta$) show it does not force RH.

C159.7 *DH "ghost-sensitive by construction".* Partly corrected. The signed coefficients $(1, \kappa, -\kappa, -1, 0)$ do break every gas positivity, but that makes the layer DH-sensitive, not ghost-sensitive (§9).

C159.8 *Legendre structure.* Correct. The transform is involutive only on convex functions; otherwise it returns the convex hull (Maxwell construction).

C159.9 *PV/TS is a symplectomorphism.* Correct ([THEOREM], Thm. 4.1). It is one 2-form in two charts, it is equivalent to the four Maxwell relations, it preserves orientation, and it holds for every region and every simple system, not only cycles. Ritchie (1968) not opened.

C159.10 *"Work = cohomology".* Refined. Work is the *flux* of an exact 2-form, not a cohomology class. The cohomological content is the first law $[\delta Q - \delta W] = 0$. On the arithmetic side the periods of $d\log\Lambda$ are $2\pi i \cdot \#\text{zeros}$, and these are ghost-blind (Prop. 5.2).

C159.11 *Carathéodory.* Refined. The integrating factor of δQ always exists in 2D, so the content appears only in ≥ 3 dimensions and needs the adiabatic-inaccessibility (Planck/Kelvin) principle. Carathéodory 1909 not opened (Wikipedia "Second law", fetched).

C159.12 *Clausius / Γ as entropy production.* Corrected. $\Gamma \geq 0$ is automatic, and RH is the *equality* $\Gamma \equiv 0$ (reversibility, L118). The working positivity law is passivity, where ghosts appear as *gain*, the opposite of "ghost = dissipation". Under the forward heat flow in de Bruijn–Newman, dissipation *creates* ghosts. A zero-free entropy-production functional is [OPEN].

C159.13 *The missing compatible J.* Correct as a pointer to MP4, but in 2D J is free (Prop. 7.1(c)). The content is $[J, \Theta] = 0$. Mrugała's metric is indefinite.

C159.14 *Name collision.* "Carathéodory" in L157 Thm. 6.2 refers to the Carathéodory function class (positive real part), not to the thermodynamic Carathéodory principle (§5).

C159.15 *Refinement of L116 (S1).* Real-axis convexity is unconditional only for positive weights. DH has $(\log \mathrm{DH})'' < 0$ on $(1, 4.2318318633607)$. The property is DH-sensitive, not ghost-sensitive.

The thirteen chat claims are C159.1–13. C159.14–15 are new audit items found during this build.

11. Ranked directions, questions, atlas, sources

Directions (score $= 0.6\,\mathrm{sig} + 0.4\,\mathrm{prob}$; all [HEURISTIC] estimates).

ID	direction	sig	prob	score
T1	Sector-wise gas/KMS test: run the ghost ladder per $\hat{\mathbb{Z}}^*$ sector (BC $\beta > 1$ states) to see whether a sector carries the FE coupling	.55	.80	.65
T4	Equation-of-state ghost detection: $P_X(\beta)$ for DH vs ζ near $\beta = \frac{1}{2}$ (link to L155)	.45	.85	.61
T5	What separates ζ from Epstein/$\zeta\zeta$ in gas language: FE self-duality + infinite divisibility + degree one	.65	.55	.61
T2	Passivity margin from primes (`mp_passivity_margin`)	.75	.35	.59
T3	A zero-free entropy-production functional that is not Γ	.70	.30	.54
T6	Contact/Reeb flow commuting with Ruppeiner J (Reeb flows and Deninger systems mutually exclusive on closed M, earlier result)	.70	.15	.48

Questions for J.W.

Q1 Adopt passivity as the precise "second law" reading, with Clausius/Γ kept as an [ANALOGY]?

Q2 Accept C159.2 (no volume; P_X as the equation of state) and C159.5 (Fisher, not Lee–Yang)?

Q3 Record t727u as Theorem `pv_ts_symplectomorphism`, with "work is a flux" (C159.10) as the reading of "work = cohomology"?

Q4 Next step: T1, or T4/T5?

Q5 Fold the L159 delta (together with pending L157 and L158) into `concepts/atlas.json`, giving 204 concepts?

Atlas delta (not folded; `build159/atlas_delta.json`). `build159/atlas_delta.json`, scored by `concepts/scores.py` on a merged copy: `concepts/atlas.json` v1.1 (188 concepts, already containing L155/L156) plus the *pending, unfolded* L157 delta (193) and L158 delta (198) plus this delta. It is **not** folded into `concepts/`. Existing entries were checked first (`kms_bost_connes`, `debruijn_newman`, `legendre_support_function`, `passivity`, `epstein`, `half_density`, `archimedes`); there was no primon, Lee–Yang, Ruppeiner or contact entry. New entries in descending priority: `free_gas_positivity_ladder` 0.54 (rank

61); `lee_yang_fisher_rh` 0.54 (rank 67); `arithmetic_equation_of_state` 0.42 (rank 95); `clausius_gamma_reversibility` 0.392 (rank 104); `pv_ts_symplectomorphism` 0.307 (rank 162); `thermo_geometry_ruppeiner_contact` 0.307 (rank 164). Updated: `davenport_heilbronn`, `passivity`, `debruijn_newman`, `epstein`, `legendre_support_function`, `gamma_fwhm_of_a_zero`, `kms_bost_connes`, `half_density`, `archimedes`. Merged copy: 204 concepts, 625 relations.

Sources opened this build (fetched 2026-10-27 to `lit/lemma159/`): Wikipedia "Maxwell relations", "Clausius theorem", "Second law of thermodynamics", "Grand canonical ensemble", "Foster's reactance theorem", "Entropy production", "Hurwitz zeta function", "Thermodynamic square", "Conjugate variables (thermodynamics)", "Thermodynamic equilibrium", "Ruppeiner geometry", "Laws of thermodynamics". **Local copies re-read**: Bost–Connes 1995 (`lit/lemma31/bc95.txt`); Wikipedia "Primon gas", "Lee–Yang theorem", "Lee–Yang theory" (`lit/lemma124/src`); Newman–Wu, arXiv:1901.06596; Mrugała 2000 (`lit/lemma116/src`). Earlier TAA builds L83, L103, L104, L116, L118, L124, L150, L153, L155–L158 (read, untouched). Certified data: L156 widths, `num/l104/zeros.json`.

Not opened (cited only through the sources above, or not used): Ritchie 1968 (PV/TS); Carathéodory 1909; Lee–Yang 1952 originals; Julia 1990 and Spector 1990; Ruppeiner, *Rev. Mod. Phys.* 1995; Weinhold 1975; Gibbs, Hermann (through Mrugała); Davenport–Heilbronn 1936 and Cassels 1961 (through Wikipedia "Hurwitz"); Foster 1924; Rodgers–Tao (arXiv 2018; Forum Math. Pi 8, 2020) and Polymath 2019 (through L124 copies, not re-read here); Lagarias and Hinkkanen on passivity (through L118).